SE Lab 7: Fuzzing for Input Validation Bugs

Problem 37 – Irrigation Scheduler (irrigation_scheduler.py)

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```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north
Traceback (most recent call last):
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 17, in main
    result = parse_zone(user_input)
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 2, in parse_zone
    name, area = text.split(':')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Figure 1: parse_zone: input "north" (no ":" separator) crashes with ValueError (not enough values to unpack).

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,,50
Traceback (most recent call last):
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 22, in main
    result = moisture_average(user_input)
  File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 6, in moisture_average
    values = [int(r) for r in readings_csv.split(',')]
              ~~~^^^
ValueError: invalid literal for int() with base 10: ''
```

Figure 2: moisture_average: input "40,,50" (empty element) crashes with ValueError (invalid literal for int()).

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): nan|8
Water Needed: nan
```

Figure 3: water_needed: input "nan|8" does not crash but silently returns nan.

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python -m pytest test_irrigation_scheduler.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 16 items

test_irrigation_scheduler.py FFFFFFF.F.FF.FF.                            [100%]

================================== FAILURES ===================================
_____________________________ test_parse_zone_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 17, in test_parse_zone_no_crash
  |     @given(st.text() | st.none())
  |                      ^^^
  |   File "C:\Users\stuth\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 20, in test_parse_zone_no_crash
    |     parse_zone(s)
    |     ^^^^^^^^^^^^^
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 3, in parse_zone
    |     return name.strip().title(), float(area)
    |                                  ^^^^^^^^^^^
```

Figure 4: Test session start: 16 tests collected; most fuzz tests fail (F).

```
_____________________________ test_moisture_average_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 26, in test_moisture_average_no_crash
  |     @given(st.text() | st.none())
  |                      ^^^
  |   File "C:\Users\stuth\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 29, in test_moisture_average_no_crash
    |     moisture_average(s)
    |     ^^^^^^^^^^^^^^^^^^^
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 6, in moisture_average
    |     values = [int(r) for r in readings_csv.split(',')]
    |                  ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 31, in test_moisture_average_no_crash
    |     assert False, f"moisture_average crashed with: {type(e).__name__}: {e}"
    | AssertionError: moisture_average crashed with: ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_moisture_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py:30
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 29, in test_moisture_average_no_crash
    |     moisture_average(s)
    |     ^^^^^^^^^^^^^^^^^^^
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\irrigation_scheduler.py", line 6, in moisture_average
    |     values = [int(r) for r in readings_csv.split(',')]
    |                             ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7\test_irrigation_scheduler.py", line 31, in test_moisture_average_no_crash
    |     assert False, f"moisture_average crashed with: {type(e).__name__}: {e}"
    | AssertionError: moisture_average crashed with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_moisture_average_no_crash(
    |     s=None,
```

Figure 5: Hypothesis failure report for test_moisture_average_no_crash: minimal counterexamples s="" and s=None.

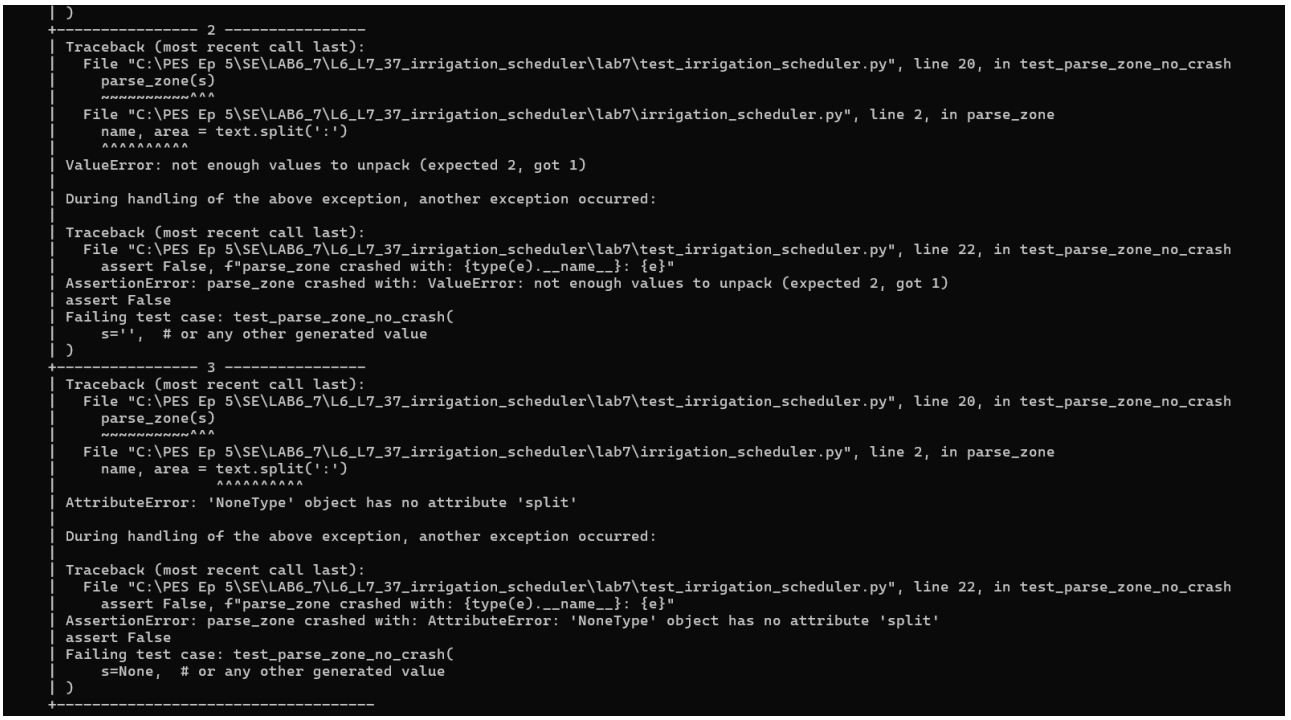


Figure 6: Hypothesis failure report for test_parse_zone_no_crash: counterexamples s="" and s=None.

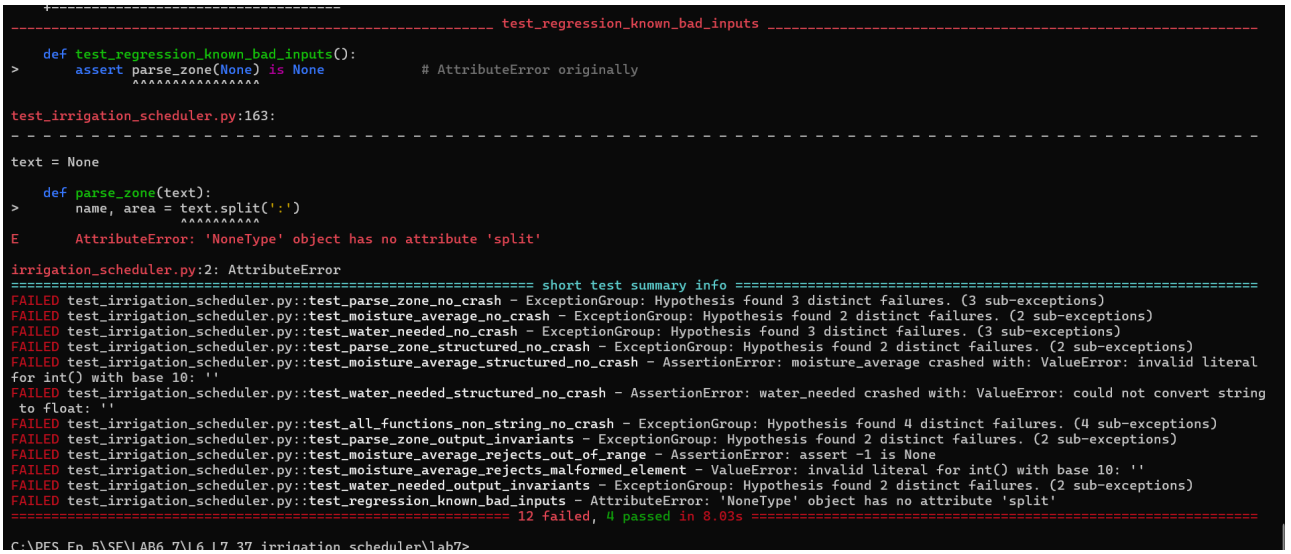


Figure 7: Short test summary: 12 failed, 4 passed.

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north
Zone and Area: Invalid input (expected name:area, area > 0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 120|8
Water Needed: 592.0
```

Figure 8: Fixed parse_zone: "north" now prints an invalid-input message instead of crashing.

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,,50
Average Moisture: Invalid input (expected integers 0-100, comma-separated)

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 120|8
Water Needed: 592.0

C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): nan|8
Water Needed: Invalid input (expected area|rainfall, both >= 0)
```

Figure 9: Fixed moisture_average and water_needed: "40,,50" and "nan|8" are both rejected with clear messages (two separate runs).

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 120|8
Water Needed: 592.0
```

Figure 10: Fixed script with valid input: results are unchanged from the original (North, 120.0; 41; 592.0).

```
C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7>python -m pytest test_irrigation_scheduler.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\stuth\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\PES Ep 5\SE\LAB6_7\L6_L7_37_irrigation_scheduler\lab7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 16 items

test_irrigation_scheduler.py::test_parse_zone_no_crash PASSED                                                                      [  6%]
test_irrigation_scheduler.py::test_moisture_average_no_crash PASSED                                                                [ 12%]
test_irrigation_scheduler.py::test_water_needed_no_crash PASSED                                                                    [ 18%]
test_irrigation_scheduler.py::test_parse_zone_structured_no_crash PASSED                                                           [ 25%]
test_irrigation_scheduler.py::test_moisture_average_structured_no_crash PASSED                                                     [ 31%]
test_irrigation_scheduler.py::test_water_needed_structured_no_crash PASSED                                                         [ 37%]
test_irrigation_scheduler.py::test_all_functions_non_string_no_crash PASSED                                                        [ 43%]
test_irrigation_scheduler.py::test_parse_zone_valid_roundtrip PASSED                                                               [ 50%]
test_irrigation_scheduler.py::test_parse_zone_output_invariants PASSED                                                             [ 56%]
test_irrigation_scheduler.py::test_moisture_average_within_bounds PASSED                                                           [ 62%]
test_irrigation_scheduler.py::test_moisture_average_rejects_out_of_range PASSED                                                    [ 68%]
test_irrigation_scheduler.py::test_moisture_average_rejects_malformed_element PASSED                                               [ 75%]
test_irrigation_scheduler.py::test_water_needed_non_negative_and_finite PASSED                                                     [ 81%]
test_irrigation_scheduler.py::test_water_needed_output_invariants PASSED                                                           [ 87%]
test_irrigation_scheduler.py::test_regression_known_bad_inputs PASSED                                                              [ 93%]
test_irrigation_scheduler.py::test_regression_valid_inputs_unchanged PASSED                                                        [100%]

============================================================ 16 passed in 5.58s =============================================================
```

Figure 11: All 16 tests pass against the fixed script (16 passed in 5.58s).

-